



Machine Learning Task: HR Attrition Prediction (Classification)



Problem Statement

Employee attrition is a critical issue for organizations, leading to increased hiring costs, productivity loss, and reduced employee morale.

In this project, the goal is to build a **Machine Learning classification model** that predicts whether an employee will leave the company (Attrition: Yes/No) based on features such as:

- Age
- Salary
- Job Role
- Job Satisfaction
- Work Experience
- Overtime
- Work-Life Balance
- Department

Students are required to analyze the dataset, preprocess the data, perform feature engineering, train multiple classification models, and evaluate their performance to identify the best model for predicting employee attrition.



Objective

The objective of this task is to:

- Understand and explore the HR dataset
- Perform data preprocessing (missing values, duplicates, outliers)
- Conduct Exploratory Data Analysis (EDA)
- Apply encoding and feature scaling
- Train multiple classification algorithms
- Evaluate models using classification metrics
- Perform cross-validation and hyperparameter tuning
- Select the best-performing model



Tasks

1. Import Required Libraries

2. Load the Dataset

3. Check Dataset

- Shape
- Columns
- Data types
- Summary statistics

4. Handle Missing Values

5. Remove Duplicate Records

6. Perform Univariate Analysis

- Distribution of numerical features
- Count plots for categorical features

7. Perform Bivariate Analysis

- Attrition vs categorical features
- Attrition vs numerical features

8. Generate Correlation Heatmap

9. Detect and Handle Outliers

10. Apply Encoding

- Label Encoding
- One-Hot Encoding

11. Apply Scaling

- StandardScaler / MinMaxScaler

12. Separate Features and Target Variable

- $X \rightarrow$ Features
- $y \rightarrow$ Attrition

13. Perform Train-Test Split

14. Balance data : SMOTE

15. Train Classification Models

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- KNN Classifier

16. Evaluate Models Using

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix

17. Perform Cross Validation

18. Apply Hyperparameter Tuning

- GridSearchCV
- RandomizedSearchCV

19. Compare Model Performance

- Create comparison table
- Identify best model

20. Select Best Model

- Based on metrics
- Business interpretation

21. Final Insights

- Key factors influencing attrition
- Business recommendations